Bloom Filter-Based MPSI

Weekly Progress Meeting 12 March 2026

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Bibliography

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[3] Ou Ruan, Changwang Yan, Jing Zhou, and Chaohao Ai. A practical multiparty private set intersection protocol based on bloom filters for unbalanced scenarios. Applied Sciences, 13(24):13215, 2023.

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Progress

From the plan last week:

- Do some writing on the mitigations
- Continue on the authorized

Progress

APSI 1 – Busy Judge

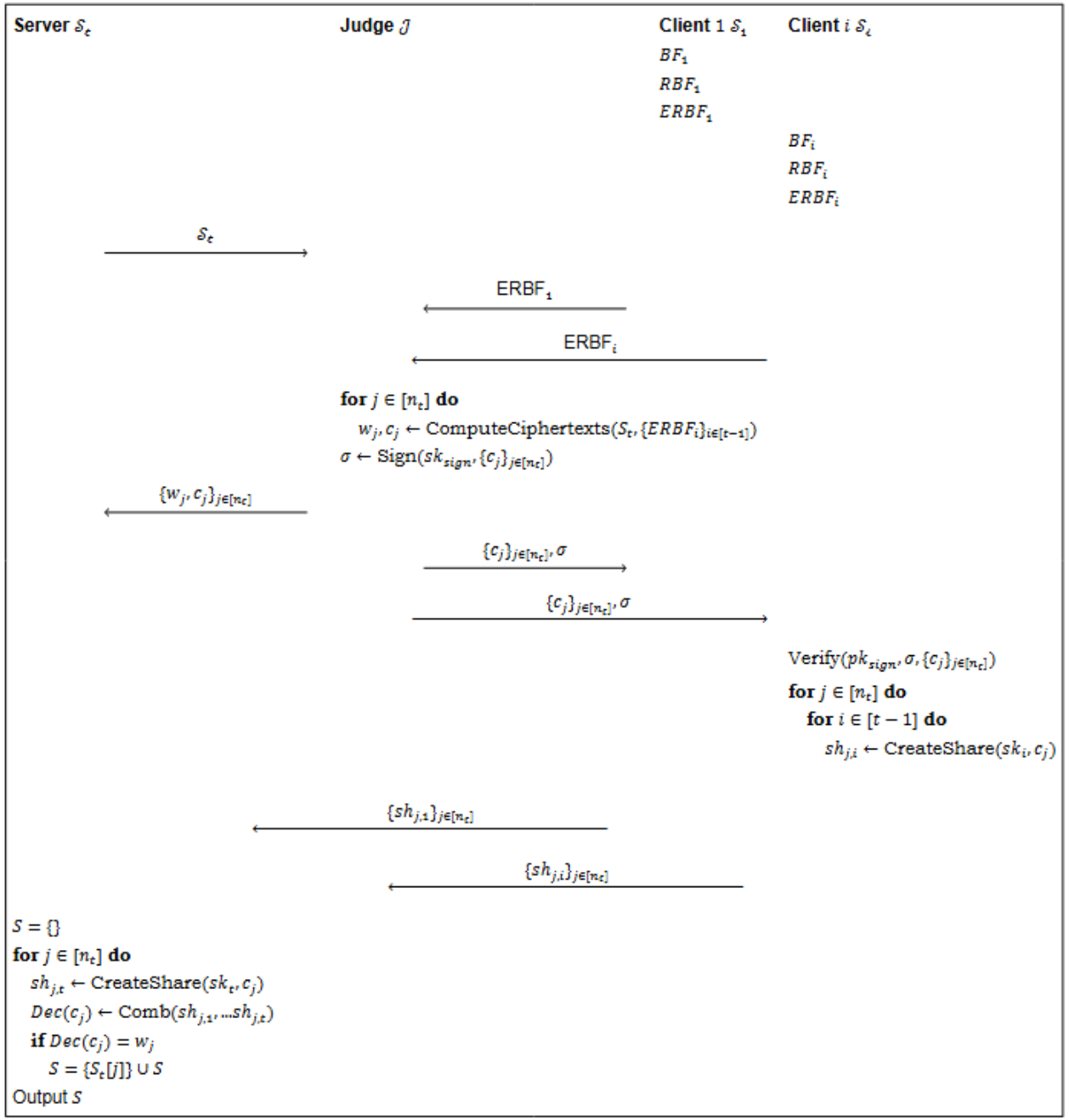
$\boldsymbol{n_t}$ = server's set size

$\boldsymbol{m}$ = bin count (BF size)

$\boldsymbol{k}$ = number of hash functions

$\boldsymbol{t}$ = number of parties (including the server)

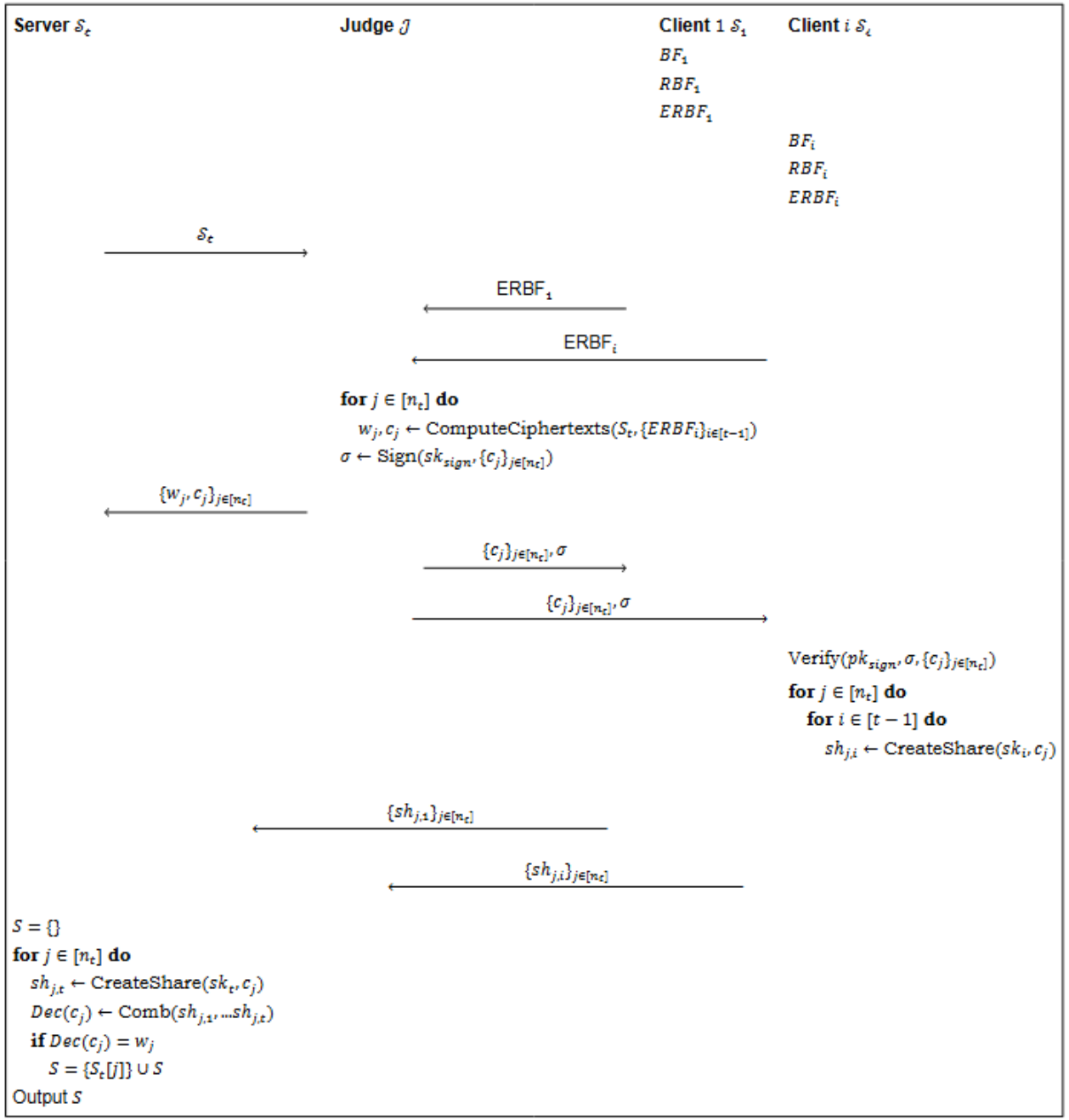
$\boldsymbol{ERBF_i}$ = the encrypted randomized Bloom filter of party i



Progress

APSI 1 – Busy Judge

- Similar to Ruan et al. [2], but a lot of the computation is “outsourced” to the Judge
- The Judge receives the Encrypted Randomized Bloom Filters (ERBFs) from all clients
- For each element of the server, the Judge aggregates all bins that the element hashes to from all ERBFs and also adds a blinded version of this element into the aggregation
- The clients jointly decrypt, the server combines the shares and adds the element to the intersection if the decryption is the blinded element

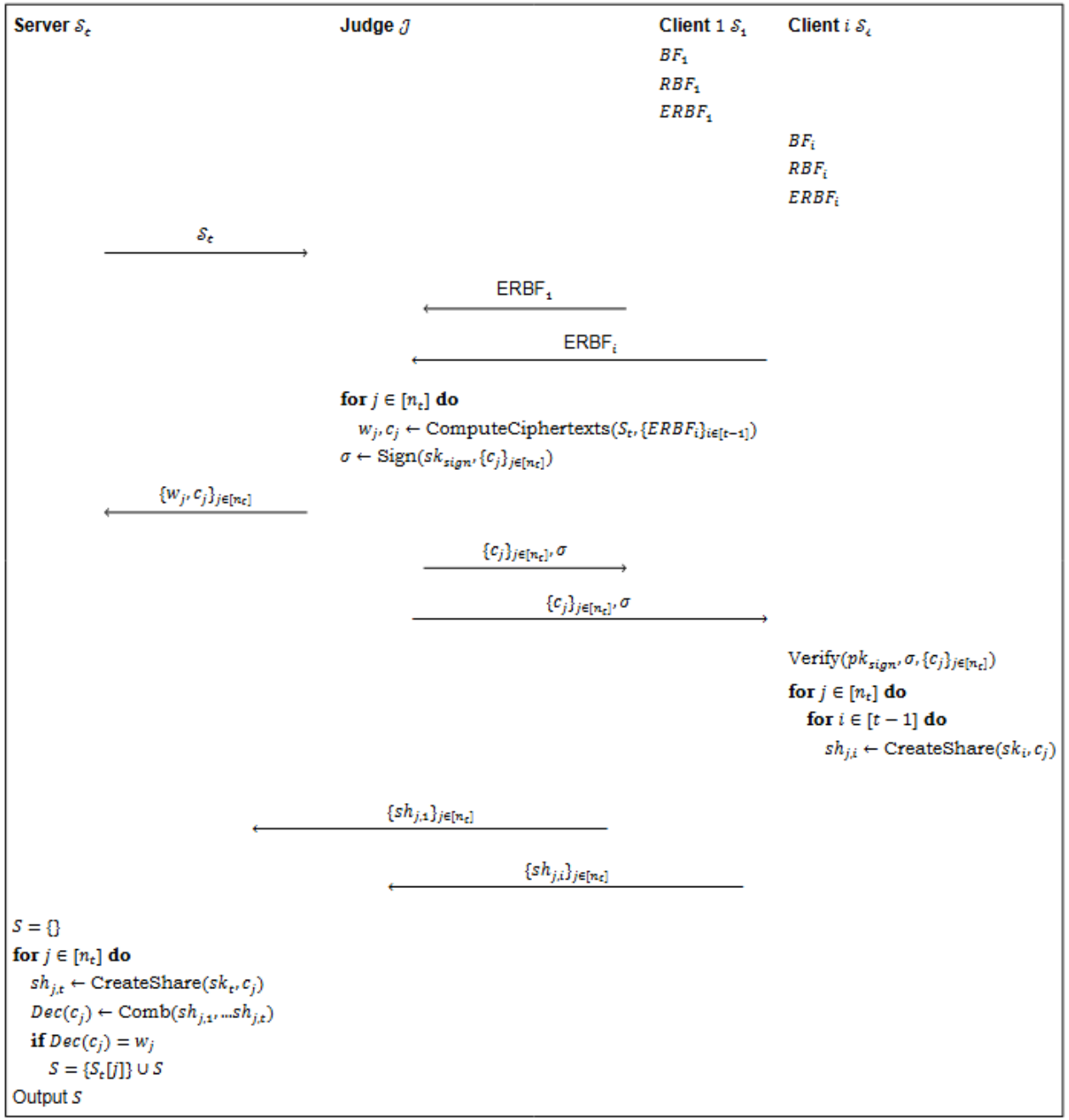


Progress

APSI 1 – Busy Judge

+ Server doesn't have to know the hash functions
+ Server can be malicious

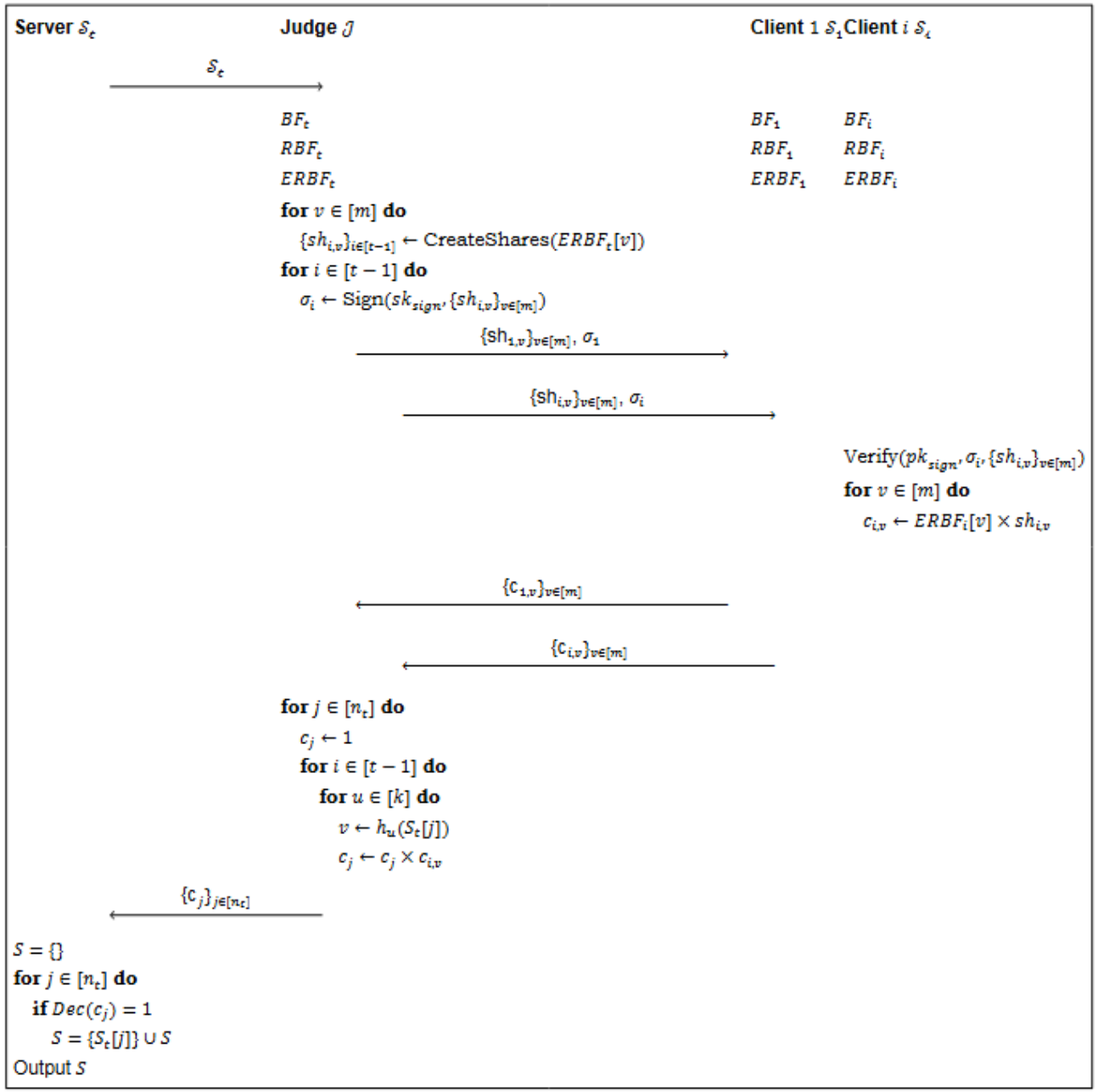
- Judge does a lot of computation
- There is communication overhead



Progress

APSI 2 – Interactive Judge

- The Judge receives the server's elements and computes an ERBF with them, then creates shares for each bin of the ERBF
- The Judge sends each client their shares, to which the clients add, for each bin, the bin of their own ERBF, send this ciphertext back to the Judge
- For each element of the server, the Judge aggregates all bins that the element hashes to from all ciphertexts, sends these aggregations to the server
- The server decrypts, adds the element to the intersection if the decryption is 1

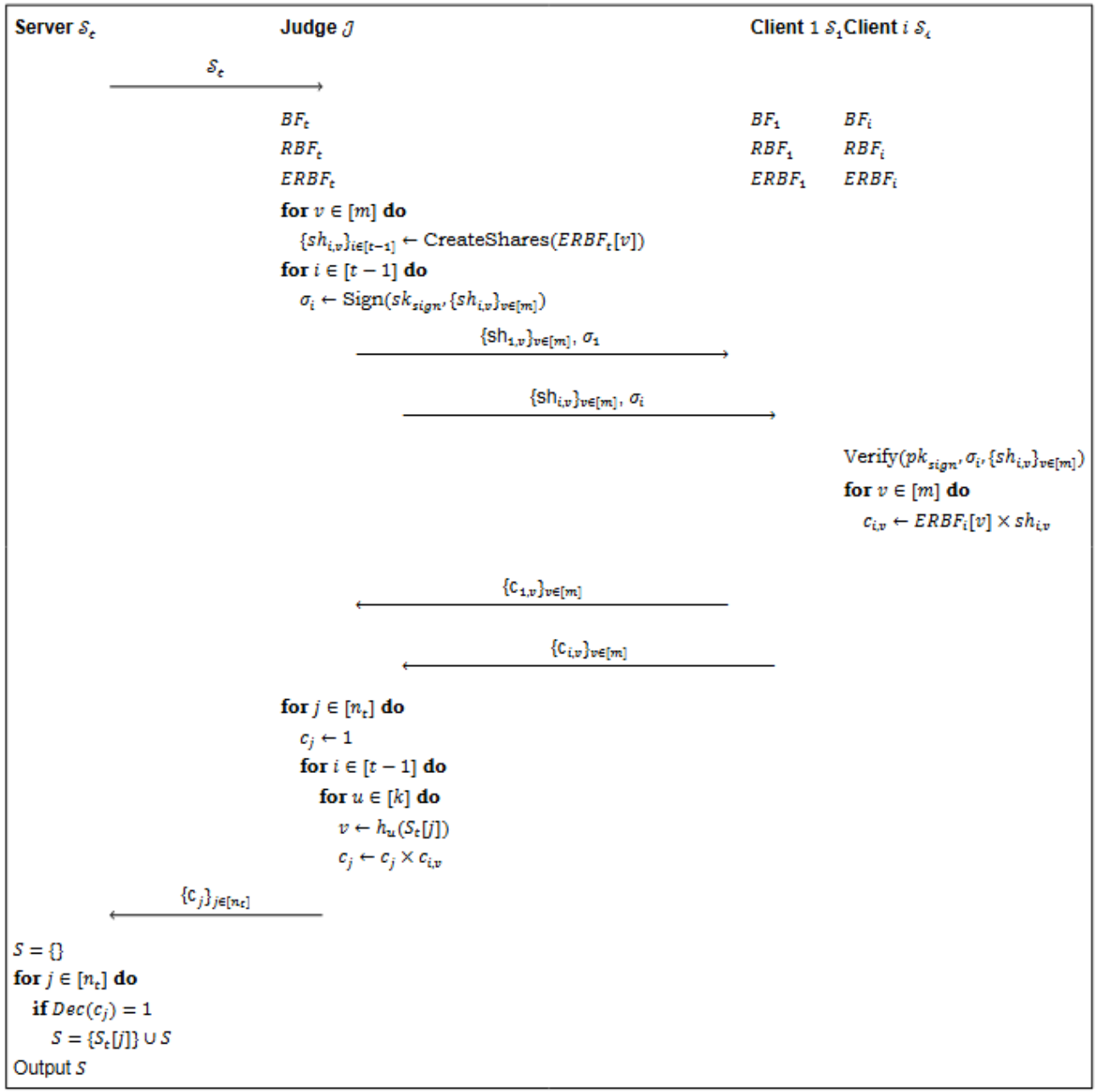


Progress

APSI 2 – Interactive Judge

+ Server doesn't have to know the hash functions
+ Communication overhead less than APSI 1
+ Server can be malicious

- Judge has to wait on computation from clients



Next Week

- Implement APSI1, then start implementing APSI2
- Do some writing

Thank you!

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